

Table 43. EMI characteristics for $f_{HSE} = 16$ MHz and $f_{HCLK} = 144$ MHz

Symbol	Parameter	Conditions	Monitored frequency band	Value	Unit
S_{EMI}	Peak ⁽¹⁾	$V_{DD} = 3.6$ V, $T_A = 25$ ° C, LQFP100 package compliant with IEC 61967-2	0.1 MHz to 30 MHz	16	dBμV
			30 MHz to 130 MHz	-1	
			130 MHz to 1 GHz	21	
			1 GHz to 2 GHz	13	
	Level ⁽²⁾		0.1 MHz to 2 GHz	3.5	-

1. Refer to the EMI radiated test section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).
2. Refer to the EMI level classification section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).

5.3.12 Electrical sensitivity characteristics

Based on three different tests (ESD, latch-up) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts × (n+1) supply pins). This test conforms to the ANSI/JEDEC standard.

Table 44. ESD absolute maximum ratings

Specified by design and not tested in production.

Symbol	Ratings	Conditions	Packages	Class	Maximum value ⁽¹⁾	Unit
V _{ESD(HBM)}	Electrostatic discharge voltage (human body model)	T _A = 25°C conforming to ANSI/ESDA/JEDEC JS-001	LQFP100	2	3400 ⁽²⁾	V
			LQFP80			
			LQFP64			
			LQFP64 alternative pinout	3A	4000	
			LQFP48	2	3400 ⁽²⁾	
			UFQFPN48			
			LQFP32			
			UFQFPN32			
V _{ESD(CDM)}	Electrostatic discharge voltage (charge device model)	T _A = 25°C conforming to ANSI/ESDA/JEDEC JS-002	LQFP100	C3	1000	
			LQFP80		1000	
			LQFP64		1250	
			LQFP64 alternative pinout		1250	
			LQFP48		1250	
			UFQFPN48		1250	
			LQFP32		1250	
			UFQFPN32		1250	

1. Evaluated by characterization - Not tested in production.
2. All pins are characterized at a maximum value of 4000 V, except for PC14, whose maximum value is 3400 V, and PC15, whose maximum value is 3600 V.